BUSINESS REQUIREMENTS DOCUMENT

Vindhyan Studio

E-commerce platform for custom 3D printed products

Lithophane Photos | Key Rings | 3D Person Models | Small Statues
Mechanical Calendars | Bathroom Premium Kits | Custom Flower Pots | Premium Lamps

Version	1.0 (Draft for review)
Date	11 October 2026
Prepared for	Vindhyan Studio - Business Owner
Status	Awaiting approval

1. Executive Summary

Vindhyan Studio will sell made-to-order and ready-stock 3D printed products online across India. The store must let a customer browse a catalogue, personalise a product (photo, name or text), sign in with a **mobile number or email**, pay through an **online payment gateway**, receive an automatic **tax invoice** and have the order shipped through **Shiprocket**. The owner manages the catalogue by editing one **Excel sheet** and one **photo folder**, with no developer needed.

Business goal	Success measure
Launch a self-running online store	Orders flow from payment to shipment with no manual re-typing
Make catalogue updates easy	A new SKU appears on the site within a few seconds of adding it to Excel and photos
Reduce friction at checkout	Login by OTP or email; checkout completed in under 2 minutes
Build customer trust	Instant invoice, order tracking, clear shipping and return rules

2. Scope

In scope	Out of scope (for version 1)
Public storefront and product pages Login with mobile OTP or email Profile page (details, addresses, orders, invoices) Customisation inputs and photo upload Cart, wishlist, checkout, online payment and COD Auto invoice after payment Shiprocket shipping and tracking Excel and photo-folder catalogue sync Basic owner admin view of orders	Native mobile apps Marketplace or multi-seller features International shipping and multi-currency Live chat, loyalty points and referral programs Automated 3D slicing or print scheduling Accounting software integration

User roles: Customer (browses, buys, tracks), **Owner / Admin** (updates catalogue, processes orders), and external systems: **Payment gateway**, **Shiprocket** and an **SMS/OTP provider**.

3. Product Catalogue

Category	What the customer buys	Customer input needed
Lithophane Photos	Thin 3D printed panel that shows a photo when lit from behind (frames, night lights)	Upload 1 photo; choose size and shape
Key Rings	Personalised key rings	Name or text; colour
3D Person Model	Small figurine or bust made from photos of a person	Upload 2-3 photos; height and pose notes
Small Statues	Decorative statues and figurines	Optional colour / size choice
Mechanical Calendar	Printed desk calendar with a mechanical date mechanism	Colour; optional name or logo
Bathroom Premium Kits	Matching sets for the bathroom (holders, hooks, organisers)	Kit variant and colour
Flower Pots (Custom)	Printed pots with custom text, pattern or size	Text or pattern; size; colour
Premium Lamps	Designer table and night lamps	Variant, colour; optional text or photo

Every product is a **SKU**. Existing Excel columns stay the same: *SKU_id, Product_Name, Product_category, MRP, Selling_Price, %offer, stock, descriptions*. Photos for a SKU live in the folder *photos/<SKU_id>/* and all images in that folder appear on the product page.

4. Functional Requirements

4.1 Login, Registration and Profile

ID	Requirement	Priority
FR-01	Customer can register and log in with a **mobile number + OTP** or with **email + password / email OTP**.	Must
FR-02	Both methods lead to the same account; a customer can add the second method later in the profile.	Must
FR-03	OTP is 6 digits, valid for 5 minutes, resend after 30 seconds, limited attempts to prevent abuse.	Must
FR-04	**Profile page**: name, mobile, email, profile photo, change password, logout.	Must
FR-05	Profile shows **saved addresses** (add, edit, delete, set default), **order history** with status, and **invoice downloads**.	Must
FR-06	Profile shows wishlist shortcut and a "Reorder" button on past orders.	Should

4.2 Catalogue, Product, Cart and Wishlist

ID	Requirement	Priority
FR-07	Catalogue is read from the Excel sheet; photos are matched to a SKU by folder name. New or changed rows appear without a code change.	Must
FR-08	Home page: category browsing, search, price and offer display (MRP, selling price, % off), stock badge ("Only 3 left", "Out of stock").	Must
FR-09	Product page: photo gallery of all images, description, price, stock, customisation fields, Add to Cart and Wishlist.	Must
FR-10	Cart: change quantity, remove items, live totals including shipping. Wishlist works for guests and syncs after login.	Must
FR-11	Price and stock are always taken from the server, never from the customer browser.	Must

4.3 Customisation

ID	Requirement	Priority
FR-12	Products flagged as customisable ask for input: photo upload (JPG/PNG, size limit), name/text, colour, size.	Must
FR-13	Customisation details and uploaded photos are saved with the order line and visible to the owner for production.	Must
FR-14	Customer sees a clear note that customised products are made to order, with the estimated production time.	Should

4.4 Checkout, Payment and Invoice

ID	Requirement	Priority
FR-15	Checkout requires login, a delivery address and a payment method (online or COD, COD configurable).	Must
FR-16	**Payment gateway** (Razorpay recommended) supports UPI, cards, netbanking and wallets. Payment is confirmed only after the server verifies the gateway signature.	Must
FR-17	Failed or abandoned payments keep the order as "Payment pending" and allow retry; no stock or shipment is created until payment succeeds.	Must
FR-18	After successful payment the system **automatically generates a tax invoice (PDF)** with a unique number, and makes it available on the success page, in Profile and by email/SMS link.	Must
FR-19	Invoice contains: seller name, address and GSTIN, invoice number and date, customer name and address, item list with HSN code, taxable value, GST split, shipping, discount, total and payment mode.	Must
FR-20	Refunds and cancellations update the order and generate a credit note.	Should

4.5 Shipping (Shiprocket)

ID	Requirement	Priority
FR-21	Pincode check on the cart or checkout page shows whether delivery is available and the expected delivery date.	Must
FR-22	On successful payment (or COD confirmation) the order is created in **Shiprocket** automatically with address, items, weight and size.	Must
FR-23	Courier, AWB number and tracking link are saved to the order; the customer sees live status in Profile and receives SMS/email updates.	Must
FR-24	Shiprocket status updates (picked up, in transit, delivered, RTO) are received by webhook and shown on the order.	Should
FR-25	Shipping charge rule is configurable (for example free above a set order value, flat fee below it).	Must

4.6 Owner / Admin

ID	Requirement	Priority
FR-26	Owner updates catalogue through Excel and photo folders only; the system reports rows with errors (missing price, duplicate SKU, no photo).	Must
FR-27	Owner views orders, customisation files and invoices, and can mark production status (Printing, Ready, Dispatched).	Must
FR-28	Stock is reduced when an order is paid, and reported back so the Excel stock figure can be refreshed.	Should

5. Integrations

System	Purpose	Key steps
Payment gateway (Razorpay)	Collect online payments securely	1. Server creates gateway order from the verified cart total 2. Customer pays in the gateway pop-up 3. Server verifies signature and marks order Paid 4. Webhook handles late success, failure and refunds

System	Purpose	Key steps
Shiprocket	Courier selection, labels, tracking	1. Authenticate with API user 2. Check pincode serviceability 3. Create order after payment 4. Assign courier and get AWB 5. Receive tracking updates by webhook
SMS / OTP provider	Mobile login and order messages	Send and verify OTP; send order, shipping and invoice link messages (provider to be selected, for example MSG91 or Twilio)
Email service	Email OTP, invoice and order mails	Transactional email provider such as SMTP or SendGrid

6. Business Rules, Non-Functional Needs and Risks

Area	Requirement
Business rules	Prices are tax-inclusive unless the owner decides otherwise. Free shipping threshold and COD are configurable. Customised items are non-returnable except for damage or print defects (policy to be confirmed).
Security	HTTPS only, hashed passwords, OTP rate limits, no card data stored on our server (handled by the gateway), uploads checked for type and size.
Performance	Pages load in under 3 seconds on mobile networks; catalogue handles at least 500 SKUs and 5 photos per SKU.
Usability	Mobile-first design; works on current Chrome, Safari and Edge; clear error messages.
Compliance	GST-compliant invoice, Privacy Policy, Terms, Shipping and Refund pages, consent for storing customer photos.
Availability	Target 99% uptime; daily backup of orders, users and uploaded files.

Risk / Assumption	Mitigation
Wrong or missing data in Excel (price, duplicate SKU)	Validation report on every sync; bad rows are skipped, not shown
Customer photos are low quality for lithophane or 3D models	Photo guidance on the product page and owner approval before printing
OTP SMS costs and delivery delays	Offer email login as fallback; choose a DLT-registered provider
Courier delays or RTO on COD	Pincode check, COD limit for high-value orders, SMS tracking updates

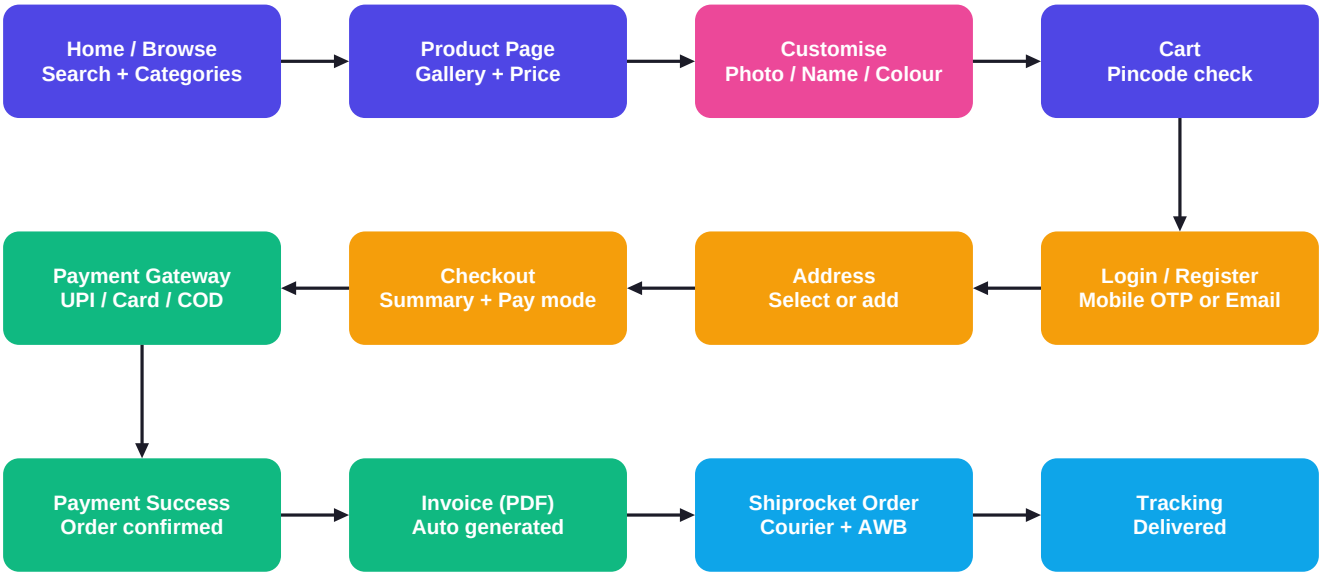
7. Current Prototype vs This BRD

A working prototype already exists (Excel and photo-folder catalogue, cart, wishlist, email login, addresses, Razorpay and Shiprocket hooks). The items below are still to be built.

Feature	Status
Excel + photo-folder catalogue, cart, wishlist, address book, order creation	Done in prototype
Razorpay payment with signature check, Shiprocket order creation	Done in prototype (needs live keys and testing)
Mobile number + OTP login	To build
Full profile page (details, photo, invoices, reorder)	To build (basic account page exists)
Customisation fields and photo upload per product	To build
Auto PDF tax invoice after payment	To build
Pincode check, courier tracking, Shiprocket webhooks	To build
Admin order screen, stock deduction, Excel validation report	To build

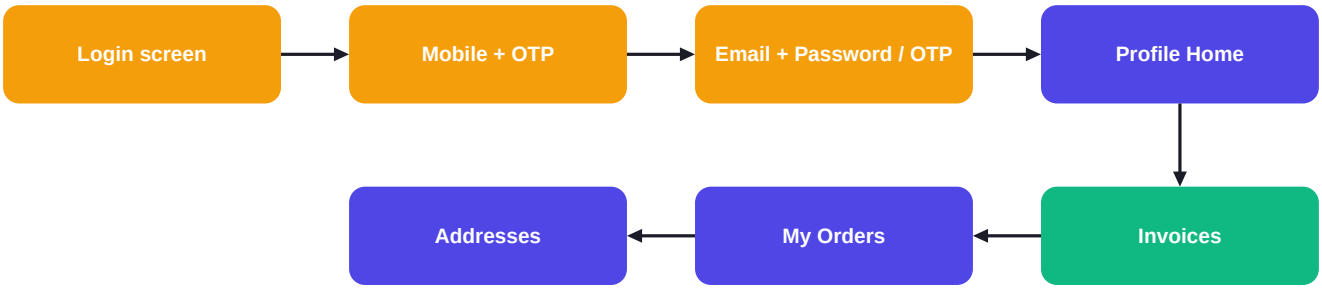
8. Screen Flow

8.1 Customer journey



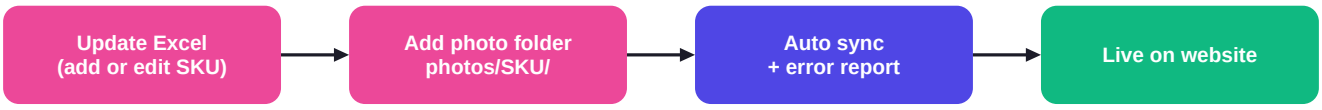
Colour key: purple = shopping, pink = customisation, amber = account and checkout, green = payment and invoice, blue = shipping. A logged-in customer skips the login step. Wishlist is reachable from Home, Product and Profile.

8.2 Login and Profile



The customer may choose either login method on the same screen; both open the same Profile. Profile links to Addresses, Orders (with tracking and Reorder) and Invoices.

8.3 Owner data flow



8.4 Screen list

#	Screen	Main elements
S1	Home / Catalogue	Categories, search, product cards, offer badges
S2	Product page	Photo gallery, price, stock, customisation form, Add to Cart / Wishlist
S3	Cart	Items, quantity, pincode check, shipping and total
S4	Login / Register	Mobile OTP and email options, resend OTP, terms consent
S5	Profile	Details, photo, addresses, orders, invoices, logout
S6	Wishlist	Saved products, move to cart
S7	Checkout and Payment	Address, payment mode, summary, gateway pop-up
S8	Order Success and Invoice	Order number, invoice download, expected delivery
S9	Order Tracking	Status timeline, courier, AWB link
S10	Owner Admin	Orders, customisation files, production status, catalogue errors

Open points for approval: seller GSTIN and HSN codes, OTP provider, COD policy, return and refund rules, production lead time per category.